

AGONPROJECT

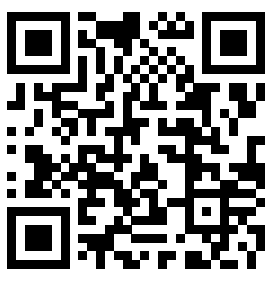
An online platform for exploring approaches to formal argumentation



WEB PLATFORM • OPEN SOURCE

The whole zoo of formal argumentation, in one browser tab.

Formal argumentation is a rich but fragmented field. AgonProject gives students, researchers and practitioners a single, **visual** interface to construct, evaluate and compare its many formalisms.

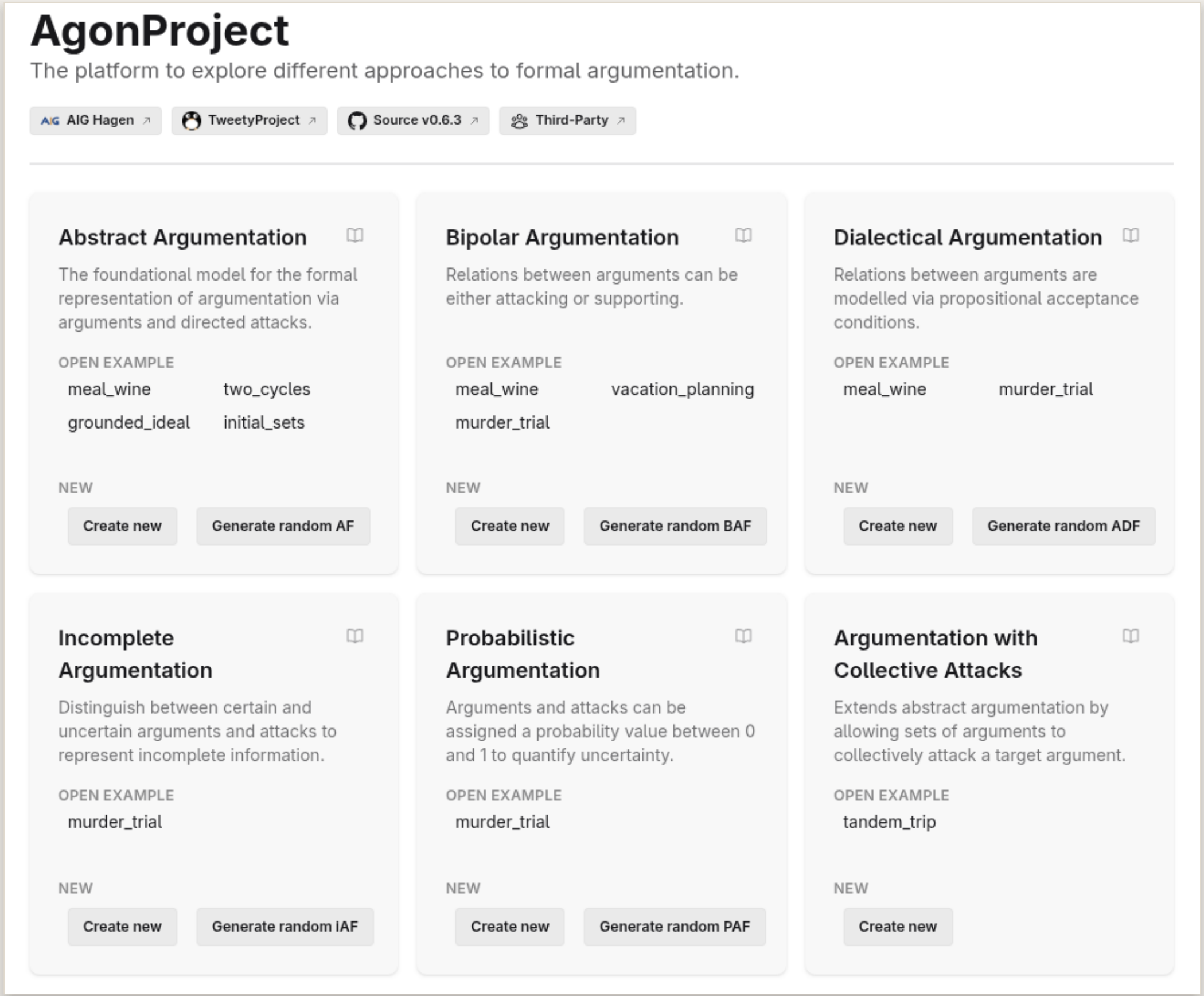


TRY IT LIVE

agon.tweetyproject.org

github.com/aig-hagen/AgonProject • GPL-3.0

 **POWERED BY TweetyProject** — reasoning libraries & solvers



Landing page — pick a formalism and start modelling, with curated examples and one-click random generation.

FEATURES

Construct

Interactive graph editor with automatic layouts and physics simulation.



Evaluate

Extension- and ranking-based semantics; enumeration, credulous / skeptical acceptance.



Export & share

LaTeX, SVG, TGF and ICCMA AF-format, plus share-URLs.



Generate

Random frameworks from eight configurable networkX generators.



Learn

10+ tutorials and a glossary with definitions and literature references.



IN ACTION

Extension & Ranking Semantics

Extensions: Preferred · Enumerate

Semantics: Preferred Mode: Enumerate

Preferred Semantics ⓘ
A set of arguments E is a preferred extension iff E is complete and $\subseteq$ -maximal.

Evaluate ☒ Evaluate continuously

Preferred extensions
{Fish, Red} {Fish, White} {Meat, Red}
{Pasta, Red}

Select extension to highlight: - 1ms

Copy results

Ranking: Categorizer

Semantics: Categorizer

Categorizer Ranking ⓘ
The categorizer ranking associates to any AF F a ranking $\succeq_F^C$ on A such that $\forall a, b \in A: a \succeq_F^C b$ iff $\text{Cat}_F(a) \geq \text{Cat}_F(b)$, where Cat_F is the categorizer function.

Evaluate ☒ Evaluate continuously

Categorizer ranking
Pasta 0.5 Fish 0.5 Meat 0.5 White 0.37 Red 0.73

Node labels show ranking scores. - 1ms

Copy ranking

Preferred extensions enumerated, with the categoriser ranking annotated on each argument.

Collective attacks & Exporting

Export

Format: LaTeX (argumentation)

Style Parameters

```
\usepackage[namemy]{amsmath}[argumentation]
```

Save text Copy text Save SVG Copy SVG

```
1 \begin{af}
2 \argument{a1}{a} at (1.00,1.73)
3 \argument{a2}{b} at (2.00,0.00)
4 \argument{a3}{c} at (3.00,1.73)
5 \argument{a4}{d} at (4.00,0.00)
6 \argument{a5}{e} at (4.50,0.87)
7 \argument{a6}{f} at (0.00,0.00)
8 \setattach(a1,a2){a3}
9 \setattach(a3,a4,a5){a2}
10 \setattach(a1,a4){a2}
11 \end{af}
```

Set-attacks drawn as hyperedges; the export panel emits ready-to-use LaTeX with an SVG preview.

ADFs & Highlighting

Models: Stable · Enumerate

Parameters

Semantics: Stable Mode: Enumerate

Stable Semantics (ADF) ⓘ
A three-valued interpretation v of an ADF is a stable model iff $v \models^T(t) = v_{\text{pr}}^T(t)$, where v_{pr} is the grounded model of the reduced ADF D_{pr} .

Evaluate ☒ Evaluate continuously

Stable models
{Guilty, ~Innocent, Motive, ~Alibi, Evidence}
{~Guilty, Innocent, Motive, Alibi, ~Evidence}

Select model to highlight: - 1ms

Copy results

ADFs attach propositional conditions to arguments; a stable model is highlighted in the graph.

Fourth panel

Placeholder — drop the fourth screenshot here and replace with `\shotpanel`.

SUPPORTED SEMANTICS

Classical	conflict-free, admissible, complete, grounded, preferred, stable
Admissibility-based	ideal, semi-stable, eager, strong admissibility, initial sets, ...
Non-admissible	naive, CF2, SCF2, stage, stage2, (strongly) undisputed
Meta-Families	weak admissibility, (semi-)qualified, serialisable
Rankings	burden, categoriser, counting, discussion, iterated graded defense, serialisability, ...

OUTLOOK

- more formalisms — ABA, weighted AFs, BSAFs, claim-augmented AFs
- benchmark generators from the ICCMA ecosystem
- explanation-based reasoning: *why* is an argument accepted?

AgonProject is backend-agnostic and open to new formalisms and solvers.

agon.tweetyproject.org • github.com/aig-hagen/AgonProject • powered by  TweetyProject

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